Artificial Intelligence - Coursework 2

Conversational Academic Advisor Agent

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Domain Description

The domain I chose for this coursework is a small academic advisor agent that helps assess whether a student may be at low, medium, or high academic risk. I chose this domain because it is narrow, clearly defined, and safe, which fits the coursework requirements well. The system is not meant to replace a tutor or make official academic decisions. Its purpose is only to take a small amount of academic information from the user and give a simple risk assessment with recommendations.

The University of Roehampton provides a particularly relevant real-world context for this kind of tool. Roehampton is a widening participation institution in southwest London, with a diverse student body of over 11,000 students drawn from more than 140 countries [4]. A significant proportion of its students come from non-traditional backgrounds, including mature students, care-experienced students, and those from lower socioeconomic groups who may face additional pressures beyond their academic workload. Nationally, the share of UK undergraduate students reporting mental health difficulties rose from 6% to 16% between 2016 and 2023, with financial stress and academic pressure cited as leading causes [5]. In this context, an early academic risk tool could act as a lightweight first line of support, helping students at Roehampton self-identify warning signs before they escalate into more serious problems that require intervention from tutors or student wellbeing staff.

The agent works using four main factors: attendance, coursework score, deadline proximity, and subject difficulty. These were chosen because they are directly connected to academic progress and are simple enough to model in a small conversational system. A student with low attendance, weak coursework, and a close deadline is more likely to be struggling than a student with good attendance, strong coursework, and enough time before the next deadline.

Student Context and Problems Addressed

The problems this agent addresses are grounded in real challenges faced by students in UK higher education. Three main issues are relevant. First, attendance decline is often an early sign of disengagement. Students may miss lectures due to work commitments, mental health difficulties, or personal pressures, but may not realise how quickly this affects their academic standing until it is too late. At Roehampton, timetables are structured across no more than three days per week to acknowledge students' wider commitments [4], but attendance can still fall without students seeking support.

Second, coursework performance is a direct indicator of academic understanding. A student who consistently scores below 50 on coursework may be struggling with the material, but without a prompt to seek help, this pattern may go unnoticed until formal assessment. Research on early warning systems in higher education consistently shows that performance data combined with engagement data is among the most effective combination of signals for identifying students at risk [6].

Third, deadline management is a practical problem that many students struggle with. Students who leave work until the final few days face higher stress, lower quality output, and a greater risk of submission failure. This is especially relevant for students who are balancing part-time work or caring responsibilities alongside their studies.

The agent targets these three problems by asking for structured input and returning a personalised risk classification with concrete recommendations. Rather than replacing a tutor or wellbeing advisor, the tool is intended to act as a prompt: a low-effort, always-available first step that encourages students to reflect on their situation and seek support before problems become serious. This fits within Roehampton's existing support model, which includes college-based wellbeing officers and pastoral support staff [4], by helping students identify when they should reach out to those services.

Knowledge Representation

The system stores knowledge mainly in the form of rules, risk labels, Bayesian priors, and recommendations. The rule-based part connects certain academic conditions to risk levels. For example, attendance below 50% is treated as high risk, coursework between 50 and 64 is treated as medium risk, and a deadline within 3 days is treated as urgent. These rules form the main knowledge base of the agent.

The system also stores prior probabilities for high, medium, and low risk. These are later updated using the evidence given by the user. In addition, the system stores a list of recommendations for each risk level. For high risk, the advice is more urgent, such as booking a meeting with a tutor. For medium risk, the advice is more about improving consistency. For low risk, the advice focuses on maintaining the current study strategy. This means that the knowledge representation does not only include rules for deciding risk, but also actions the agent should suggest after making a decision.

Inference Method

The system uses two inference methods: forward chaining and Bayesian reasoning. Forward chaining is used in the rule-based part. This means the agent starts from the facts given by the user and checks which rules can be triggered. For example, if the attendance is below 50, the low attendance rule is triggered. If the deadline is within 3 days, the urgent deadline rule is triggered. In this way, the agent moves from facts to conclusions.

The second method is Bayesian reasoning. The agent starts with prior probabilities for high, medium, and low risk, and then updates them depending on the evidence given by the user. For example, low attendance increases the probability of high risk, while strong coursework increases the probability of low risk. The final result is based on the highest Bayesian probability. Using both methods together makes the system stronger: the rules make the reasoning easy to explain, while the Bayesian part helps the system reason under uncertainty.

Uncertainty Handling

Uncertainty handling is an important part of the system because users do not always provide full information. The first way the system handles uncertainty is through Bayesian probabilities. Instead of only giving a fixed label, the agent calculates probabilities for high, medium, and low risk. This means the final answer is not treated as absolute certainty, but as the most likely outcome based on the available evidence.

The second way uncertainty is handled is through missing data. The user can type “skip” or “I don’t know” when answering questions. The program validates input and does not crash when this happens. If too little information is given, the system clearly says that it does not have enough information for a useful assessment. This is important because it is safer than pretending to know the answer when the evidence is too limited.

The system also explains what data was used, what rules were triggered, and what Bayesian probabilities were produced. This makes the uncertainty visible to the user and improves transparency.

Evaluation Examples

Case 1 - High Risk

```
[*]: if __name__ == "__main__":
    agent = AcademicAdvisorAgent()
    agent.run()
```

```
==============================================
      ACADEMIC ADVISOR INTELLIGENT AGENT
==============================================
Welcome! I am your AI Academic Advisor.
I am designed to help you assess your current academic
risk level and provide study recommendations.

You can start by describing your situation, for example:
'I have 40% attendance' or 'My coursework score was 55'.

Type 'exit' to end our session.
==============================================

You:  i have low attendance and my deadline is in 2 days

Please answer these questions. You can press Enter to keep a suggested value or type 'skip' if unsure.

Attendance percentage (0-100):  [45]:
Coursework score (0-100):  48
Days until deadline (0-365):  [2]:
Difficulty (easy / medium / hard):  hard

[System] Analyzing data using Rule-based and Bayesian logic...

Decision: high_risk
Recommendations:
- Book an urgent meeting with your tutor
- Make a daily revision plan
- Speak to your lecturer about weak areas

Explanation:
- I used this information: {'attendance': 45, 'coursework': 48, 'deadline': 2, 'difficulty': 'hard'}
- The rule-based part suggests: high_risk
- The Bayesian part gives these probabilities: high=0.591, medium=0.269, low=0.14
- So the final answer is high_risk because it has the highest probability.
- Confidence in this result: 0.591
- Rules that were triggered:
  * Attendance is below 50%
  * Coursework score is below 50
  * Deadline is very close (3 days or less)
  * The subject feels hard
- This is only a study-support tool, not an official academic judgement.
--------------------------------------------------

Is there anything else you'd like to discuss?
You:
```

The first screenshot shows a student saying they have low attendance and that the deadline is in 2 days. The agent suggested attendance as 45 and deadline as 2, then after the remaining inputs were given it classified the student as high risk. This makes sense because attendance is below 50, coursework is below 50, the deadline is very close, and the subject is hard. This is a successful case because both the rule-based part and the Bayesian part clearly point to high risk.

Case 2 - Medium Risk

```
[*]: if __name__ == "__main__":
    agent = AcademicAdvisorAgent()
    agent.run()
```

```
==============================================
      ACADEMIC ADVISOR INTELLIGENT AGENT
==============================================
Welcome! I am your AI Academic Advisor.
I am designed to help you assess your current academic
risk level and provide study recommendations.

You can start by describing your situation, for example:
'I have 40% attendance' or 'My coursework score was 55'.

Type 'exit' to end our session.
==============================================

You:  my attendance is 67 and the subject feels hard

Please answer these questions. You can press Enter to keep a suggested value or type 'skip' if unsure.

Attendance percentage (0-100):  [67]:
Coursework score (0-100):  72
Days until deadline (0-365):  14
Difficulty (easy / medium / hard):  [hard]:  hard

[System] Analyzing data using Rule-based and Bayesian logic...

Decision: medium_risk
Recommendations:
- Revise every week
- Use past papers
- Join or form a study group

Explanation:
- I used this information: {'attendance': 67, 'coursework': 72, 'deadline': 14, 'difficulty': 'hard'}
- The rule-based part suggests: medium_risk
- The Bayesian part gives these probabilities: high=0.219, medium=0.446, low=0.335
- So the final answer is medium_risk because it has the highest probability.
- Confidence in this result: 0.446
- Rules that were triggered:
  * Attendance is between 50% and 69%
  * The subject feels hard
- This is only a study-support tool, not an official academic judgement.
--------------------------------------------------

Is there anything else you'd like to discuss?
You:  ↑↓ for history. Search history with c-↑/c-↓
```

The second screenshot shows a student whose attendance is 67 and who says the subject feels hard. The coursework is strong and the deadline is not close, so not all indicators point to serious difficulty. In this case the system classified the student as medium risk. This is expected because the student has some warning signs, but not enough for a high-risk judgement. This case is useful because it shows that the system can make a balanced decision rather than always jumping to the most severe result.

Case 3 - Low Risk

```
[*]: if __name__ == "__main__":
    agent = AcademicAdvisorAgent()
    agent.run()
```

```
==============================================
      ACADEMIC ADVISOR INTELLIGENT AGENT
==============================================
Welcome! I am your AI Academic Advisor.
I am designed to help you assess your current academic
risk level and provide study recommendations.

You can start by describing your situation, for example:
'I have 40% attendance' or 'My coursework score was 55'.

Type 'exit' to end our session.
==============================================

You:  i am doing fine

Please answer these questions. You can press Enter to keep a suggested value or type 'skip' if unsure.

Attendance percentage (0-100):  85
Coursework score (0-100):  78
Days until deadline (0-365):  20
Difficulty (easy / medium / hard):  [easy]:  easy

[System] Analyzing data using Rule-based and Bayesian logic...

Decision: low_risk
Recommendations:
- Keep using your current study plan
- Practise more exam-style questions

Explanation:
- I used this information: {'attendance': 85, 'coursework': 78, 'deadline': 20, 'difficulty': 'easy'}
- The rule-based part suggests: low_risk
- The Bayesian part gives these probabilities: high=0.194, medium=0.259, low=0.547
- So the final answer is low_risk because it has the highest probability.
- Confidence in this result: 0.547
- No risk rules were triggered from the rule base.
- This is only a study-support tool, not an official academic judgement.
--------------------------------------------------

Is there anything else you'd like to discuss?
You: ↑↓ for history. Search history with c-↑/c-↓
```

The third screenshot shows a more positive case where the student says they are doing fine. Strong attendance, strong coursework, a comfortable deadline, and easy subject difficulty were entered. The system classified the student as low risk. This is the correct result because none of the main factors suggest academic struggle. This case shows that the agent is able to recognise students who are doing well, not only those who are at risk.

Case 4 - Partial Information / Uncertainty

```
[*]: if __name__ == "__main__":
    agent = AcademicAdvisorAgent()
    agent.run()
```

```
==============================================
      ACADEMIC ADVISOR INTELLIGENT AGENT
==============================================
Welcome! I am your AI Academic Advisor.
I am designed to help you assess your current academic
risk level and provide study recommendations.

You can start by describing your situation, for example:
'I have 40% attendance' or 'My coursework score was 55'.

Type 'exit' to end our session.
==============================================

You:  help

Please answer these questions. You can press Enter to keep a suggested value or type 'skip' if unsure.

Attendance percentage (0-100):  skip
Coursework score (0-100):  55
Days until deadline (0-365):  skip
Difficulty (easy / medium / hard):  medium

[System] Analyzing data using Rule-based and Bayesian logic...

Decision: medium_risk
Recommendations:
- Revise every week
- Use past papers
- Join or form a study group

Explanation:
- I used this information: {'coursework': 55, 'difficulty': 'medium'}
- The rule-based part suggests: medium_risk
- The Bayesian part gives these probabilities: high=0.259, medium=0.483, low=0.259
- So the final answer is medium_risk because it has the highest probability.
- Confidence in this result: 0.483
- Rules that were triggered:
  * Coursework score is between 50 and 64
- This is only a study-support tool, not an official academic judgement.
--------------------------------------------------

Is there anything else you'd like to discuss?
You:
```

The fourth screenshot shows a case where the user only says “help”, so the system begins with very little information. Attendance and deadline were skipped, but coursework score and difficulty were provided. Based on this limited input, the system still produced a medium-risk result. This is a good example of uncertainty handling because the agent continues working with partial data instead of failing. At the same time, it only uses the information that is actually available.

Case 5 - Limitation / Failure Case

```
[*]: if __name__ == "__main__":
    agent = AcademicAdvisorAgent()
    agent.run()
```

```
===============================================
      ACADEMIC ADVISOR INTELLIGENT AGENT
===============================================
Welcome! I am your AI Academic Advisor.
I am designed to help you assess your current academic
risk level and provide study recommendations.

You can start by describing your situation, for example:
'I have 40% attendance' or 'My coursework score was 55'.

Type 'exit' to end our session.
===============================================

You:  i am stressed and behind in everything

Please answer these questions. You can press Enter to keep a suggested value or type 'skip' if unsure.

Attendance percentage (0-100):  skip
Coursework score (0-100):  skip
Days until deadline (0-365):  skip
Difficulty (easy / medium / hard):  skip

[!] More Information Needed
To provide a useful assessment, I need at least two details.
Please tell me about your attendance, coursework, or deadlines.
--------------------------------------------------
You:  ↑↓ for history. Search history with c-↑/c-↓
```

The fifth screenshot shows a user saying they are stressed and behind in everything, but then skipping all four inputs. In this case the agent says that it does not have enough information for a useful assessment. This is an important limitation example. A human advisor would understand that the user sounds worried, but the system cannot make a proper decision without structured academic information. This shows that the agent depends heavily on user-provided input and has limited natural language understanding.

Evaluation

The five examples show that the system performs reasonably well in a small academic support domain. In the first three cases it works as intended and correctly distinguishes between high risk, medium risk, and low risk. This shows that the rules, Bayesian reasoning, and recommendations are working properly in normal situations. The explanations also make the system more transparent, because the user can see what information was used and why the answer was given.

The fourth example shows that the system handles partial information better than a very basic rule-only system. Even when some inputs are missing, the agent can still make a cautious judgement if enough data is available. The validation also improves robustness because inputs such as “skip” or “I don’t know” do not crash the program.

However, the system also has weaknesses. It depends heavily on structured input and does not understand vague emotional language very well. The fifth example shows this clearly. The system also uses manually chosen priors and probability multipliers, so the Bayesian part is heuristic rather than based on real student data. Another limitation is that the rule base is small and does not include factors such as exam results, personal circumstances, or long-term academic trends.

Overall, the agent is successful for the scale of the coursework. It is simple, explainable, and safe, and it demonstrates the main AI ideas required in the brief. A useful future improvement would be to make the first user message more meaningful, expand the rule base, and make the Bayesian part more grounded.

User Validation

To check whether the system worked as intended for a real user, one informal validation test was carried out with a fellow student. The participant was not given any instructions on how the system worked. They were simply asked to interact with it naturally, starting with a short description of their situation, and then answering the follow-up questions as they would normally.

The participant described their situation as having missed several lectures and feeling behind on a piece of coursework due within a week. After providing inputs of 58% attendance, a coursework score of 52, 6 days until the deadline, and medium difficulty, the system returned a medium-risk classification with recommendations to revise weekly, use past papers, and join a study group. The participant said the result felt accurate and that the recommendations were useful and practical. They also noted that the explanation section, which showed which rules had been triggered and what the Bayesian probabilities were, made the output feel more trustworthy than a simple label would have.

One piece of feedback was that the initial question prompt could be clearer. The participant initially typed only "I need help" and was then asked to provide additional details. They found this slightly confusing at first, which confirms the limitation identified in Case 5 of the evaluation. A future improvement would be to make the opening prompt more specific, for example by asking directly about attendance and coursework in the first message rather than waiting for structured follow-up questions.

Overall, this small validation test suggests the system produces results that feel reasonable to a real student and that the explainability features improve user trust. However, a more rigorous validation with a larger group of participants, using structured feedback forms and comparison against tutor assessments, would be needed before drawing stronger conclusions.

Ethical Reflection

There are several ethical issues and limitations connected to this system. First, the agent simplifies a student's situation into only four factors: attendance, coursework score, deadline, and subject difficulty. In reality, academic struggles are often more complex than that. A student may be affected by stress, illness, family issues, financial pressure, or other personal circumstances. UNESCO's guidance on AI in education stresses the importance of human oversight in educational settings, especially where outcomes may affect learners in meaningful ways [1]. This means the system should only be used as a support tool and not as a replacement for tutors or student support staff.

Another issue is bias. Even though the system is small and rule-based, the rules and Bayesian probabilities still reflect my own design choices. I chose the thresholds for low attendance, average coursework, and urgent deadlines, and I also chose the Bayesian priors and multipliers. Because of this, the system can still reflect bias even without machine learning. The ICO guidance highlights fairness and explainability as important issues in AI systems, while NIST also includes fairness, transparency, and accountability as key characteristics of trustworthy AI [2], [3]. In my case, this matters because different students may not fit neatly into the assumptions built into the system.

There is also a risk of over-trust. Since the agent gives a decision, probabilities, and recommendations, some users may think the output is more reliable than it really is. For this reason, I made the system clearly state that it is only a study-support tool and not an official academic judgement. This is important because transparency helps users understand the limits of the system.

Finally, privacy should also be considered. Even though this coursework version does not store real student records, academic information can still be sensitive. In a real university setting, privacy and data protection would need to be handled much more carefully [2], [3]. Overall, the ethical position of this system is that it should support, not replace, human judgement.

Project Management

To ensure the agent was developed effectively and met all the module requirements before the April 8th deadline, a structured project management approach was used. This helped in balancing the technical coding of the Bayesian logic with the academic writing and ethical research required by the brief.

Project Timeline (Gantt Chart)

Figure 1 shows the Gantt chart used to plan the development lifecycle. The project was split into four main stages: Planning, Development, Evaluation, and Documentation. A large portion of time was specifically allocated to "Inference & Uncertainty" because implementing the Bayesian math and the forward-chaining rules required the most care. This timeline ensured that the system was fully tested with five evaluation cases well before the final submission date.

Gantt Chart						
TO DO	Dates					
	13 Apr 2026	14 Apr 2026	15 Apr 2026	18 Apr 2026	19 Apr 2026	26 Apr 2026
Gantt Chart / Project Plan						
Ethical Reflection Section						
Domain Definition						
Forward Chaining Implementation						
Bayesian Logic & Normalization						
5 Evaluation Cases						
IEEE Formatting (Citations)						
Reviewing Bayesian Math in Report						
Final Proofreading						
Moodle Submission (.pdf & .ipynb)						

Figure 1: Gantt Chart

Task Tracking (Kanban Board)

Figure 2 shows the Kanban board used to track the daily progress of the coursework. This was particularly helpful for managing the "Explainability" and "Uncertainty" requirements. By moving tasks from the "Backlog" to "Done," it was easier to see which parts of the agent were finished and which parts, like the IEEE formatting and the final proofreading, still needed work. This iterative process ensured that no part of the assessment brief was missed.

Kanban Board

To Do (Backlog)	In Progress	Done	Time (mir
Final Proofreading	IEEE Formatting (Citations)	Domain Definition	60
Moodle Submission (.pdf & .ipynb)	Reviewing Bayesian Math in Report	Forward Chaining Implementation	60
		Bayesian Logic & Normalization	40
		5 Evaluation Cases	30
		Ethical Reflection Section	30
		Gantt Chart / Project Plan	20

Figure 2: Kanban Board

AI Use Declaration

Generative AI was used as a support tool during this coursework for debugging, improving code structure, and helping refine the written report. However, I made sure to understand the final code, the reasoning methods used, and the content included in the submission. The final work reflects my own understanding of the system and its design.

References

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